

Sample Midterm 2, MATH 4211/6211 Optimization

NAME: _____ **ID#:** _____ **Section: 4211/6211**

Show all your work. Answers without solution steps will not receive credits.

1. (4 points) Consider the problem of minimizing $f(x) = x^{4/3} = (\sqrt[3]{x})^4$, $x \in \mathbb{R}$. Note that 0 is the global minimizer of f .
 - (a) Write down the algorithm for Newton's method applied to this problem.
 - (b) Show that as long as the starting point is not 0, the algorithm in part (a) does not converge to 0 (no matter how close to 0 we start).

2. (4 points) Suppose we apply a fixed step size gradient algorithm to minimize

$$f(\mathbf{x}) = \mathbf{x}^T \begin{bmatrix} 3/2 & 2 \\ 0 & 3/2 \end{bmatrix} \mathbf{x} + \mathbf{x}^T \begin{bmatrix} 3 \\ -1 \end{bmatrix} - 22$$

- (a) Find the range of values of the step size for which the algorithm converges to the minimizer.
- (b) Suppose we use a step size of 1000 (which is too large). Find an initial condition that will cause the algorithm to diverge (not converge).

3. (4 points) Given $f : \mathbb{R}^n \rightarrow \mathbb{R}$, consider the general iterative algorithm

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha_k \mathbf{d}^{(k)},$$

where $\mathbf{d}^{(1)}, \mathbf{d}^{(2)}, \dots$ are given vectors in $\mathbb{R}^n$, and α_k is chosen to minimize $f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)})$; that is,

$$\alpha_k = \operatorname{argmin} f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}).$$

Show that for each k , the vector $\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}$ is orthogonal to $\nabla f(\mathbf{x}^{(k+1)})$ (assuming the gradient exists).

4. (4 points) Suppose $\mathbf{Q}$ is symmetric and positive definite. Show that if $\{\mathbf{d}^{(1)}, \dots, \mathbf{d}^{(n)}\}$ is a $\mathbf{Q}$ -conjugate set that is also orthogonal (i.e., $\mathbf{d}^{(i)T} \mathbf{d}^{(j)} = 0$ for all $i, j = 1, \dots, n$ and $i \neq j$), and $\mathbf{d}^{(i)} \neq \mathbf{0}$, $i = 1, \dots, n$, then each $\mathbf{d}^{(i)}$, $i = 1, \dots, n$, is an eigenvector of $\mathbf{Q}$.

5. (4 points) Consider the quadratic function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{x}^T \mathbf{b},$$

where $\mathbf{Q}$ is symmetric and positive definite. Let $\mathbf{D} \in \mathbb{R}^{n \times r}$ be of rank r , and $\mathbf{x}_0 \in \mathbb{R}^n$. Define the function $\phi : \mathbb{R}^r \rightarrow \mathbb{R}$ by

$$\phi(\mathbf{a}) = f(\mathbf{x}_0 + \mathbf{D} \mathbf{a}).$$

Show that ϕ is a quadratic function with a positive definite quadratic term.